17 The number of moose in Alaska at the start of year n is P_n
The number of moose in Alaska at the start of the following year is given by

$$P_{n+1} = 1.04\,(P_n - G) \qquad \text{where } G \text{ is a constant.}$$

At the beginning of 2015, there were 200 000 moose in Alaska.
At the beginning of 2016, there were 200 720 moose in Alaska.

Work out how many moose there were in Alaska at the beginning of 2017

$200720 = 1.04(200000 - G)$

1 mark
Substitute the given populations.

$200720 \div 1.04 = 200000 - G$

$193000 = 200000 - G$

$G = 200000 - 193000 = 7000$

1 mark
Isolate G.

$\text{Next population} = 1.04(200720 - 7000)$

1 mark
Apply the recurrence again.

$= 201468.8$

About 201469 moose

1 mark
Whole-number population.

(Total for Question 17 is 4 marks)